

Table 1. Jigsaw transfer results using controllers fitted and selected on RealToxicityPrompts. Values are mean $\pm$ standard error across five repetitions; TBD marks unrun model-method pairs.

	Method	Toxic continuations (%) $\downarrow$	Distinct-2 $\uparrow$	Perplexity $\downarrow$
Gemma-2-2B	Original	6.10 $\pm$ 0.28	0.613 $\pm$ 0.003	19.19 $\pm$ 0.16
	ITI	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD
	S-PID	0.52 $\pm$ 0.12	0.677 $\pm$ 0.003	18.96 $\pm$ 0.13
	A-LQR	0.00 $\pm$ 0.00	0.602 $\pm$ 0.002	28.36 $\pm$ 0.23
	H$_{\infty}$ (ours)	0.16 $\pm$ 0.07	0.665 $\pm$ 0.002	21.19 $\pm$ 0.16
Llama-3-8B	Original	TBD	TBD	TBD
	ITI	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD
	H$_{\infty}$ (ours)	TBD	TBD	TBD
Qwen-2.5-14B	Original	TBD	TBD	TBD
	ITI	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD
	H$_{\infty}$ (ours)	TBD	TBD	TBD